```python
import pandas as pd
import matplotlib as plt
import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio
import plotly.colors as colors

from plotly.colors import qualitative
pio.templates.default = "plotly_white"

data = pd.read_csv("SampleSuperstore.csv" , encoding = 'latin-1')

data.head()
```

```
   Row_ID        Order_ID  Order_Date   Ship_Date       Ship_Mode
Customer_ID  \
0       1  CA-2016-152156   11/8/2016  11/11/2016    Second Class
CG-12520
1       2  CA-2016-152156   11/8/2016  11/11/2016    Second Class
CG-12520
2       3  CA-2016-138688   6/12/2016   6/16/2016    Second Class
DV-13045
3       4  US-2015-108966  10/11/2015  10/18/2015  Standard Class
SO-20335
4       5  US-2015-108966  10/11/2015  10/18/2015  Standard Class
SO-20335

     Customer_Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Postal_Code  Region       Product_ID         Category
Sub_Category  \
0       42420   South  FUR-BO-10001798        Furniture    Bookcases

1       42420   South  FUR-CH-10000454        Furniture       Chairs

2       90036    West  OFF-LA-10000240  Office Supplies       Labels

3       33311   South  FUR-TA-10000577        Furniture       Tables

4       33311   South  OFF-ST-10000760  Office Supplies      Storage


                                        Product_Name     Sales
Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600
2
```

```
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                     Eldon Fold 'N Roll Cart System   22.3680
2

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164

[5 rows x 21 columns]
```

```
data.describe()
```

```
            Row_ID   Postal_Code         Sales     Quantity
Discount  \
count  9994.000000   9994.000000   9994.000000  9994.000000
9994.000000
mean   4997.500000  55190.379428    229.858001     3.789574
0.156203
std    2885.163629  32063.693350    623.245101     2.225110
0.206452
min       1.000000   1040.000000      0.444000     1.000000
0.000000
25%    2499.250000  23223.000000     17.280000     2.000000
0.000000
50%    4997.500000  56430.500000     54.490000     3.000000
0.200000
75%    7495.750000  90008.000000    209.940000     5.000000
0.200000
max    9994.000000  99301.000000  22638.480000    14.000000
0.800000

            Profit
count  9994.000000
mean     28.656896
std     234.260108
min   -6599.978000
25%       1.728750
50%       8.666500
75%      29.364000
max    8399.976000
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Row_ID         9994 non-null   int64  
 1   Order_ID       9994 non-null   object 
 2   Order_Date     9994 non-null   object 
 3   Ship_Date      9994 non-null   object 
 4   Ship_Mode      9994 non-null   object 
 5   Customer_ID    9994 non-null   object 
 6   Customer_Name  9994 non-null   object 
 7   Segment        9994 non-null   object 
 8   Country        9994 non-null   object 
 9   City           9994 non-null   object 
 10  State          9994 non-null   object 
 11  Postal_Code    9994 non-null   int64  
 12  Region         9994 non-null   object 
 13  Product_ID     9994 non-null   object 
 14  Category       9994 non-null   object 
 15  Sub_Category   9994 non-null   object 
 16  Product_Name   9994 non-null   object 
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64  
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

```
data['Order_Date'] = pd.to_datetime(data['Order_Date'])
data['Ship_Date'] = pd.to_datetime(data['Ship_Date'])
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype         
---  ------         --------------  -----         
 0   Row_ID         9994 non-null   int64         
 1   Order_ID       9994 non-null   object        
 2   Order_Date     9994 non-null   datetime64[ns]
 3   Ship_Date      9994 non-null   datetime64[ns]
 4   Ship_Mode      9994 non-null   object        
 5   Customer_ID    9994 non-null   object        
 6   Customer_Name  9994 non-null   object        
 7   Segment        9994 non-null   object        
 8   Country        9994 non-null   object        
 9   City           9994 non-null   object        
 10  State          9994 non-null   object        
```

```
 11  Postal_Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product_ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub_Category   9994 non-null   object
 16  Product_Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: datetime64[ns](2), float64(3), int64(3), object(13)
memory usage: 1.6+ MB
```

```
data.head()
```

```
   Row_ID        Order_ID Order_Date  Ship_Date       Ship_Mode
Customer_ID  \
0       1  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
1       2  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
2       3  CA-2016-138688 2016-06-12 2016-06-16    Second Class    DV-
13045
3       4  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335
4       5  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335

     Customer_Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
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3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
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  Postal_Code  Region       Product_ID         Category
Sub_Category  \
0       42420   South  FUR-BO-10001798        Furniture    Bookcases

1       42420   South  FUR-CH-10000454        Furniture       Chairs

2       90036    West  OFF-LA-10000240  Office Supplies       Labels

3       33311   South  FUR-TA-10000577        Furniture       Tables

4       33311   South  OFF-ST-10000760  Office Supplies      Storage


                                        Product_Name     Sales
Quantity  \
```

```
0                    Bush Somerset Collection Bookcase  261.9600
2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                       Eldon Fold 'N Roll Cart System   22.3680
2

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164

[5 rows x 21 columns]
```

```
data['Order_Month'] = data['Order_Date'].dt.month
data['Order_Year'] = data['Order_Date'].dt.year
data['Order_Day_Of_Week'] =  data['Order_Date'].dt.dayofweek

data.head()
```

```
   Row_ID        Order_ID Order_Date  Ship_Date       Ship_Mode
Customer_ID  \
0       1  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
1       2  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
2       3  CA-2016-138688 2016-06-12 2016-06-16    Second Class    DV-
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3       4  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335
4       5  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335

     Customer_Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
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3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

          Category  Sub_Category  \
0        Furniture     Bookcases
1        Furniture        Chairs
2  Office Supplies        Labels
```
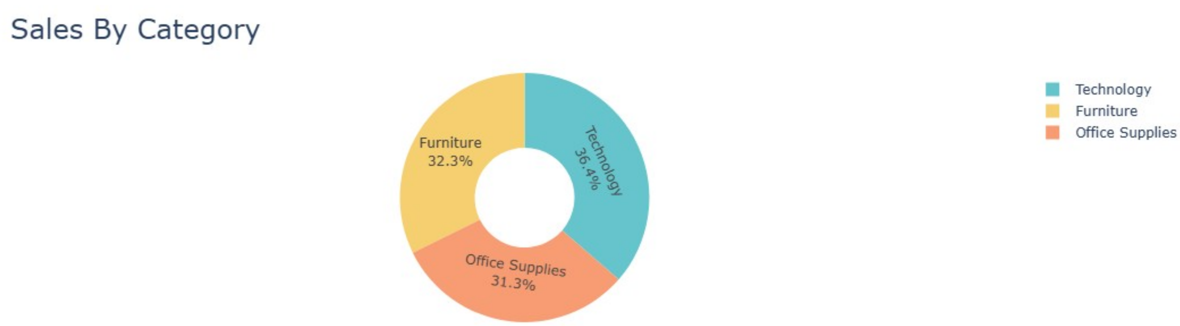

```
3        Furniture        Tables
4  Office Supplies       Storage

                                            Product_Name     Sales
Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600
2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                     Eldon Fold 'N Roll Cart System   22.3680
2

  Discount    Profit  Order_Month  Order_Year  Order_Day_Of_Week
0     0.00   41.9136           11        2016                  1
1     0.00  219.5820           11        2016                  1
2     0.00    6.8714            6        2016                  6
3     0.45 -383.0310           10        2015                  6
4     0.20    2.5164           10        2015                  6

[5 rows x 24 columns]

#monthly sales analysis

sales_by_month = data.groupby('Order_Month')
['Sales'].sum().reset_index()

sales_by_month

    Order_Month        Sales
0             1   94924.8356
1             2   59751.2514
2             3  205005.4888
3             4  137762.1286
4             5  155028.8117
5             6  152718.6793
6             7  147238.0970
7             8  159044.0630
8             9  307649.9457
9            10  200322.9847
10           11  352461.0710
11           12  325293.5035

fig = px.line(sales_by_month,
              x = 'Order_Month',
              y = 'Sales',
            title = 'Sales By Month')
fig.show()
```


```
#sales by category

sales_by_category=  data.groupby('Category')
['Sales'].sum().reset_index()

sales_by_category
```

	Category	Sales
0	Furniture	741999.7953
1	Office Supplies	719047.0320
2	Technology	836154.0330

```
fig = px.pie(sales_by_category,
             names = 'Category',
             values = 'Sales',
             hole = 0.4,
             color_discrete_sequence = px.colors.qualitative.Pastel)
fig.update_traces(textposition ='inside', textinfo='percent+label')
fig.update_layout(title_text = 'Sales By Category' , title_font =
dict(size=24))

fig.show()
```

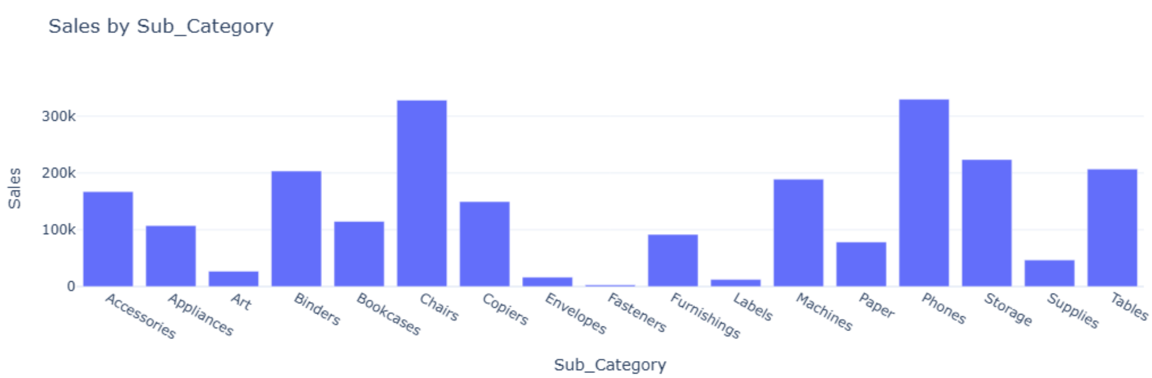



```
#sales analysis by sub-category
```

```
sales_by_sub_category=  data.groupby('Sub_Category')
['Sales'].sum().reset_index()

sales_by_sub_category
```

	Sub_Category	Sales
0	Accessories	167380.3180
1	Appliances	107532.1610
2	Art	27118.7920
3	Binders	203412.7330
4	Bookcases	114879.9963
5	Chairs	328449.1030
6	Copiers	149528.0300
7	Envelopes	16476.4020
8	Fasteners	3024.2800
9	Furnishings	91705.1640
10	Labels	12486.3120
11	Machines	189238.6310
12	Paper	78479.2060
13	Phones	330007.0540
14	Storage	223843.6080
15	Supplies	46673.5380
16	Tables	206965.5320

```
fig = px.bar(sales_by_sub_category, x = 'Sub_Category', y = 'Sales' ,
             title = 'Sales by Sub_Category')
fig.show()
```




```
#monthly profit analysis

profit_by_month = data.groupby('Order_Month')
['Profit'].sum().reset_index()

profit_by_month
```

	Order_Month	Profit
0	1	9134.4461

```
1                2   10294.6107
2                3   28594.6872
3                4   11587.4363
4                5   22411.3078
5                6   21285.7954
6                7   13832.6648
7                8   21776.9384
8                9   36857.4753
9               10   31784.0413
10              11   35468.4265
11              12   43369.1919

fig = px.bar(profit_by_month, x = 'Order_Month', y = 'Profit',
            title = 'Profit by month')
fig.show()
```

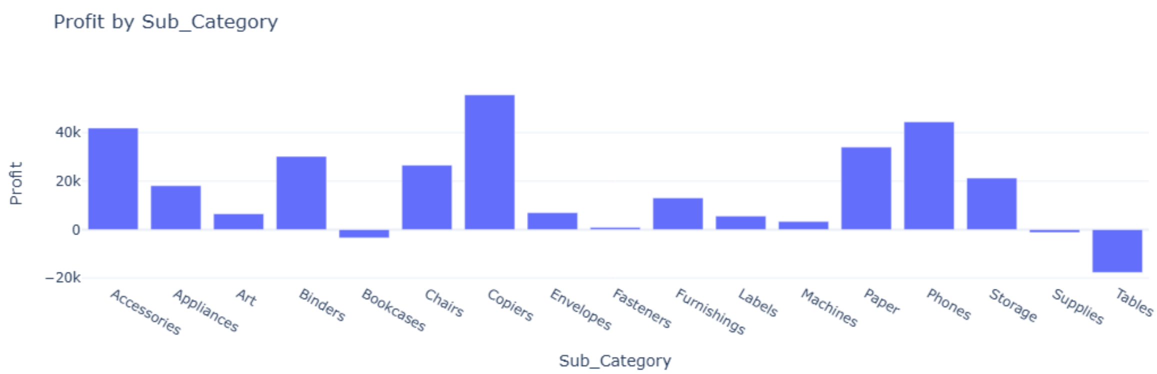

```
#profit by category

profit_by_category = data.groupby('Category')
['Profit'].sum().reset_index()

fig = px.pie(profit_by_category,
            names = 'Category',
            values = 'Profit',
            hole =  0.4,
            color_discrete_sequence = px.colors.qualitative.Pastel)
fig.update_traces(textposition = 'inside', textinfo = 'percent+label')
fig.update_layout(title_text = 'Profit by Category', title_font =
dict(size=24))
fig.show()
```

Profit by Category


```
#profit by sub_category

profit_by_sub_category = data.groupby('Sub_Category')
['Profit'].sum().reset_index()

profit_by_sub_category
```

	Sub_Category	Profit
0	Accessories	41936.6357
1	Appliances	18138.0054
2	Art	6527.7870
3	Binders	30221.7633
4	Bookcases	-3472.5560
5	Chairs	26590.1663
6	Copiers	55617.8249
7	Envelopes	6964.1767
8	Fasteners	949.5182
9	Furnishings	13059.1436
10	Labels	5546.2540
11	Machines	3384.7569
12	Paper	34053.5693
13	Phones	44515.7306
14	Storage	21278.8264
15	Supplies	-1189.0995
16	Tables	-17725.4811

```
fig = px.bar(profit_by_sub_category, x = 'Sub_Category', y = 'Profit',
title = 'Profit by Sub_Category')
fig.show()
```




```
#sale and profit by consumer segment

data.head()
```

```
   Row_ID        Order_ID Order_Date  Ship_Date       Ship_Mode
Customer_ID  \
0       1  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
1       2  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
2       3  CA-2016-138688 2016-06-12 2016-06-16    Second Class    DV-
13045
3       4  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335
4       5  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335

    Customer_Name    Segment        Country             City  ...   \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

          Category  Sub_Category  \
0        Furniture     Bookcases
1        Furniture        Chairs
2  Office Supplies        Labels
3        Furniture        Tables
4  Office Supplies       Storage

                                        Product_Name     Sales
Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600
2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
```

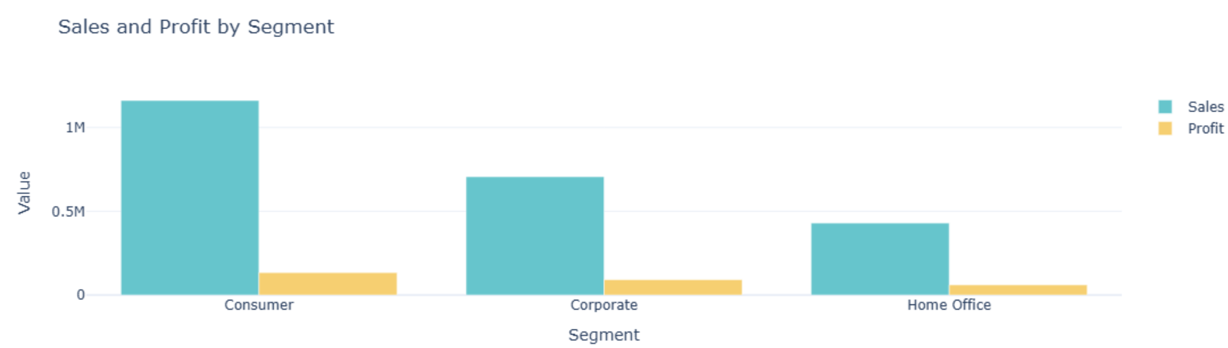
```
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                      Eldon Fold 'N Roll Cart System   22.3680
2

  Discount    Profit  Order_Month  Order_Year  Order_Day_Of_Week
0     0.00   41.9136           11        2016                  1
1     0.00  219.5820           11        2016                  1
2     0.00    6.8714            6        2016                  6
3     0.45 -383.0310           10        2015                  6
4     0.20    2.5164           10        2015                  6

[5 rows x 24 columns]
```

```
sales_profit_by_segment =
data.groupby('Segment').agg({'Sales':'sum','Profit':'sum'}).reset_inde
x()

sales_profit_by_segment
```

```
       Segment         Sales        Profit
0     Consumer  1.161401e+06  134119.2092
1    Corporate  7.061464e+05   91979.1340
2  Home Office  4.296531e+05   60298.6785
```

```
color_palette = qualitative.Pastel


fig = go.Figure()


fig.add_trace(go.Bar(
    x=sales_profit_by_segment['Segment'],
    y=sales_profit_by_segment['Sales'],
    name='Sales',
    marker_color=color_palette[0]
))


fig.add_trace(go.Bar(
    x=sales_profit_by_segment['Segment'],
    y=sales_profit_by_segment['Profit'],
    name='Profit',
    marker_color=color_palette[1]
))

# Customize layout
fig.update_layout(
    title='Sales and Profit by Segment',
```

```
    xaxis_title='Segment',
    yaxis_title='Value',
    barmode='group'
)
fig.show()
```




```
#sales profit ratio

sales_profit_by_segment =
data.groupby('Segment').agg({'Sales':'sum','Profit':'sum'}).reset_inde
x()
sales_profit_by_segment['sales_to_profit_ration'] =
sales_profit_by_segment['Sales'] / sales_profit_by_segment['Profit']
print(sales_profit_by_segment[['Segment', 'sales_to_profit_ration']])
```

```
       Segment  sales_to_profit_ration
0     Consumer                8.659471
1    Corporate                7.677245
2  Home Office                7.125416
```